```
0 REM ********************************
1 REM *                              *
2 REM *           GDEM 1.0           *
3 REM *  GRAPHICS DATA ENTRY MONITOR  *
4 REM *                              *
5 REM *        COPYRIGHT 1984        *
6 REM *                              *
7 REM *         BY DEREK LANE        *
8 REM *                              *
9 REM ********************************
10 POKE53280,6:POKE53281,6:POKE55,255:POKE56,63:CLR:POKE788,52:OPEN1,0
:OPEN2,3
15 SZ=2048:REM CHANGE THIS TO CHANGE MEMORY SIZE
20 PRINT"GDEM 1.0"
25 PRINT"GRAPHICS DATA ENTERING MONITOR"
30 CO=0:DIMKW$(10)
40 FORN=0TO8:READKW$(N):NEXT
50 DATASTOP,LIST,SAVE,LOAD,DELETE,SEARCH,CLEAR,DIR,"D:"
60 FORN=680TO715:READA:POKEN,A:NEXT
61 DATA169,2,162,8,160,255,32,186,255,169,0,162,224,160,2,32,189,255
62 DATA169,0,133,251,169,64,133,252,169,251,162,0,160,66,32,216,255,96
100 PRINT"OK."
105 PRINTMID$(STR$(CO),2);" ";:GOSUB300
110 IFLEFT$(B$,1)=">"THEN500
120 CO=VAL(B$):B$=MID$(B$,LEN(STR$(CO)))
125 IFLEFT$(B$,1)=" "THENB$=MID$(B$,2):GOTO125
130 DE$=LEFT$(B$,3):B$=MID$(B$,5)
135 IFLEFT$(B$,1)=" "THENB$=MID$(B$,2):GOTO135
140 BE=VAL(B$):B$=MID$(B$,LEN(STR$(BE))+1)
145 IFLEFT$(B$,1)=" "THENB$=MID$(B$,2):GOTO145
150 DI=VAL(B$)*10
160 L=16384+CO*7
165 A=BE:GOSUB200
170 A=DI:GOSUB200
175 FORN=1TO3:POKEL+N-1,ASC(MID$(DE$,N,1)):NEXT
180 CO=CO+1:IFCO<=SZ/7THEN105
190 ER=3:GOTO5000
200 POKEL,A AND 255
210 POKEL+1,INT(A/256)
220 L=L+2:RETURN
300 INPUT#1,A$:PRINT
310 PRINT"";:B$=""
320 GET#2,A$:IFA$<>CHR$(13)THENB$=B$+A$:GOTO320
325 IFRIGHT$(B$,1)=" "THENB$=LEFT$(B$,LEN(B$)-1):GOTO325
330 RETURN
500 REM JUMP TO SUBROUTINES
505 B$=MID$(B$,2)
510 FORN=0TO8:IFLEFT$(B$,LEN(KW$(N)))<>KW$(N)THENNEXT:ER=1:GOTO5000
515 B$=MID$(B$,LEN(KW$(N))+1)
516 IFLEFT$(B$,1)=" "THENB$=MID$(B$,2):GOTO516
520 ONN+1GOSUB550,600,1000,1100,700,900,800,1200,1300
525 IFER<>0THEN5000
530 GOTO100
550 POKE53280,14:POKE53281,6:POKE788,49:PRINT"";:END
600 IFB$=""THENSR=0:EN=SZ/7:GOTO640
610 FORN=1TOLEN(B$):IFMID$(B$,N,1)<>"-"THENNEXT:SR=VAL(B$):EN=SR-1:GOT
O640
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620 S$=LEFT$(B$,N-1):SR=VAL(S$):IFS$=""THENSR=0
630 E$=MID$(B$,N+1):EN=VAL(E$):IFE$=""THENEN=SZ/7
640 L=SR*7+16384
650 PRINTMID$(STR$(SR),2);" ";
660 PRINTCHR$(PEEK(L+4));CHR$(PEEK(L+5));CHR$(PEEK(L+6));",";
670 PRINTMID$(STR$(PEEK(L)+PEEK(L+1)*256),2);",";
680 PRINTMID$(STR$((PEEK(L+2)+PEEK(L+3)*256)/10),2)
685 GETA$:IFA$=""THEN695
690 SR=SR+1:IFSR<=ENTHEN640
695 RETURN
700 GOSUB4000
710 L=PL*7+16384+7
720 FORN=LTOSZ+16384:POKEN-7,PEEK(N):NEXT
730 RETURN
800 FORN=16384TO16384+SZ:POKEN,0:NEXT:CO=0:RETURN
900 GOSUB4000:IFER<>0THENRETURN
910 B$=STR$(PL):GOTO600
999 END
1000 E=SZ+16384:POKE709,EAND255:POKE711,INT(E/256)
1005 B$="@0:"+B$
1010 POKE690,LEN(B$)
1015 FORN=1TOLEN(B$):POKE735+N,ASC(MID$(B$,N,1)):NEXT
1020 SYS680:GOSUB4500:RETURN
1100 LOADB$,8,1:GOSUB4500:RETURN
1200 REM
1210 OPEN3,8,0,"$0":FORN=1TO3:GET#3,A$,B$:NEXT
1215 C=0
1220 IFA$<>""THENC=ASC(A$)
1225 IFB$<>""THENC=C+ASC(B$)*256
1235 GET#3,B$:IFST<>0THEN1280
1240 IFB$<>CHR$(34)THEN1235
1245 GET#3,B$:IFB$<>CHR$(34)THENPRINTB$;:GOTO1245
1250 GET#3,B$:IFB$=CHR$(32)THEN1250
1255 PRINTTAB(18);:C$=""
1260 C$=C$+B$:GET#3,B$:IFB$<>""THEN1260
1265 PRINT""LEFT$(C$,3)
1270 GETT$:IFT$=""THENRETURN
1275 IFST=0THENFORN=1TO2:GET#3,A$:NEXT:GOTO1215
1280 CLOSE3:RETURN
1300 OPEN15,8,15,B$:INPUT#15,A,B$,C,D:PRINTA;B$;C;D:CLOSE15:RETURN
4000 N=16388
4005 IFPEEK(N)=ASC(B$)THEN4030
4010 N=N+7:IFN<SZ+16384THEN4005
4020 ER=2:END
4030 FORM=0TO2:IFASC(MID$(B$,M+1))<>PEEK(N+M)THEN4010
4040 NEXT:PL=INT((N-16384)/7):RETURN
4500 OPEN15,8,15:INPUT#15,A,B$,C,D:PRINTA;B$;C;D:CLOSE15:RETURN
5000 IFER=1THENPRINT"UNKNOWN COMMAND ERROR"
5010 IFER=2THENPRINT"NONEXISTANT ENTRY"
5020 IFER=3THENPRINT"OUT OF MEMORY ERROR"
5900 ER=0:GOTO100
10000 OPEN3,8,3,"SPIRIT CITIES"
10005 GET#3,A$:PRINTA$:IFST<>0THEN10005
10010 CLOSE3

READY.
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9 ifjm=1then100
10 poke53280,6:poke53281,6:poke55,255:poke56,63:poke788,52
15 dimdi$(30):sz=4096
20 forn=12288to12327:reada:poken,a:next
30 forn=680to715:reada:poken,a:next
50 data162,0,169,0,129,251,230,251,208,250,230,252,165,252,197,253,208,240,96
51 data162,0,164,254,177,251,129,251,230,251,208,248,230,252,165,252
52 data197,253,208,240,96
60 data169,2,162,8,160,255,32,186,255,169,0,162,224,160,2,32,189,255
61 data169,0,133,251,169,64,133,252,169,251,162,0,160,66,32,216,255,96
100 print"[clr][white]gdem 1.0"
110 print"[down]graphics data entry monitor"
120 print"[down]  city editor"
125 print" dot editor"
130 print" disk utilities"
140 print" printer utilities"
150 print" exit"
160 print"[down]   move the pointer to your selection"
170 print"            and hit <return>"
180 print"[home][down 4]";:mx=5:gosub5000
190 onsegoto1000,3000,2000,4000
200 poke53280,14:print"[clr][lt blue]";:end
1000 pg=0:op=1
1100 print"[clr] delete search  clear  exit"
1105 gosub5375
1110 print" design.   range     bearing   page";pg;
1120 gosub5100
1130 print"[clr][down 2]";:mx=25:se=3
1135 gosub5005
1140 ifa$="[f1]"thenpg=pg-(1*(pg<8)):goto1100
1145 ifa$="[f7]"thenpg=pg+(1*(pg>0)):goto1100
1150 ifse=1then1500
1155 ifse=2then1400
1160 sp=1:le=5:gosub5200:de$=a$
1170 sp=11:le=5:gosub5200:di=val(a$)*10
1180 sp=21:le=3:gosub5200:be=val(a$)
1200 l=16384+pg*207+(se-3)*9
1210 pokel,beand255:pokel+1,int(be/256)
1220 pokel+2,diand255:pokel+3,int(di/256)
1230 forn=4to8:pokel+n,asc(mid$(de$,(n-3),1)):next
1240 print"[up]":goto1135
1400 printspc(35);"[rvs on]";pg;"[up]"
1410 geta$:ifa$=""then1410
1420 ifa$="[f1]"thenifpg<8thenpg=pg+1:goto1400
1430 ifa$="[f7]"thenifpg>0thenpg=pg-1:goto1400
1440 goto1100
1500 om=4
1505 geta$:ifa$=""then1500
1510 ifa$="[left]"thena=-1:gosub5300
1520 ifa$="[right]"thena=1:gosub5300
1530 ifa$="[up]"ora$="[down]"thenprint"[home]";:goto1135
1540 ifa$<>chr$(13)then1500
1550 onopgoto1600,1700,1800,1900
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```
1600 print"[home] [rvs on]exit[rvs off]      move pointer to entry"
1610 print"[home]";:gosub5005
1620 ifse=1then1100
1625 ifse=2then1100
1630 ad=16384+pg*207+(se-3)*9
1640 poke251,ad and 255:poke252,int(ad/256):poke253,72:poke254,9:sys12307
1650 goto1100
1800 print"[clr]are you sure you want to erase"
1805 input"everything? no[left 4]";yn$
1810 ifleft$(yn$,1)<>"y"then1100
1815 poke251,0:poke252,64:poke253,72:sys12288:goto1000
1900 goto100
2000 print"[clr]disk utilities"
2010 print"[down] load file
2020 print" save file
2030 print" view directory
2040 print" scratch file
2050 print" rename file
2060 print" copy file under new name
2070 print" exit
2080 print"[home][down]":mx=7:gosub5000
2090 onsegoto2100,2200,2300,2400,2500,2600,100
2100 gosub5400
2110 print"[home][down]";:mx=p:gosub5000
2120 ifse=mxthen2000
2125 print"[clr]load ";di$(se)
2126 input"correct? no[left 4]";yn$:ifleft$(yn$,1)<>"y"then2000
2130 jm=1:na$=di$(se):loaddi$(se),8,1
2140 gosub5500:goto2000
2200 e=sz+16384:poke709,eand255:poke711,int(e/256)
2205 input"[clr]file name";fi$:fi$="@0:"+fi$
2210 print"[clr]save ";mid$(fi$,4)
2211 input"correct? yes[left 5]";yn$:ifleft$(yn$,1)<>"y"then2000
2215 poke690,len(fi$)
2220 forn=1tolen(fi$):poke735+n,asc(mid$(fi$,n,1)):next
2230 sys680:print:gosub5500:goto 2000
2300 gosub5400
2310 gosub5500:goto2000
2400 gosub5400:print"[clr]";tab(27);"scratch file"
2410 print"[home][down]";:mx=p:gosub5000
2420 ifse=mxthen2000
2430 print"[clr]scratch ";di$(se)
2435 input"correct? no[left 4]";yn$:ifleft$(yn$,1)<>"y"then2400
2440 open15,8,15,"s0:"+di$(se):close15:gosub5500:goto2400
2500 gosub5400:print"[home]";tab(28);"rename file"
2510 print"[home][down]";:mx=p:gosub5000
2520 ifse=mxthen2000
2530 input"[clr]new name";n$:print"Srename ";di$(se);" as ";n$
2540 input"correct? yes[left 5]";yn$:ifleft$(yn$,1)<>"y"then2500
2550 open15,8,15,"r0:"+n$+"=0:"+di$(se):close15:gosub5500:goto2500
2600 gosub5400:print"[home]";tab(30);"copy file"
2610 print"[home][down]";:mx=p:gosub5000
2620 ifse=mxthen2000
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2630 input"[clr]new name";n$:print"[clr]copy ";di$(se);" as ";n$
2640 input"correct? yes[left 5]";yn$:ifleft$(yn$,1)<>"y"then2600
2650 open15,8,15,"c0:"+n$+"=0:"+di$(se):close15:gosub5500:goto2600
3000 pg=0:op=1
3100 print"[clr] delete  clear  exit"
3105 gosub5375
3110 print" range     bearing   page";pg;
3120 gosub5600
3130 print"[clr][down 2]";:mx=25:se=3
3135 gosub5005
3140 ifa$="[f1]"thenpg=pg-(1*(pg<21)):goto3100
3145 ifa$="[f7]"thenpg=pg+(1*(pg>0)):goto3100
3150 ifse=1then3500
3155 ifse=2then3400
3170 sp=1:le=5:gosub5200:di=val(a$)*10
3180 sp=11:le=3:gosub5200:be=val(a$)
3200 l=18432+pg*92+(se-3)*4
3210 pokel,beand255:pokel+1,int(be/256)
3220 pokel+2,diand255:pokel+3,int(di/256)
3240 print"[up]":goto3135
3400 printspc(25);"[rvs on]";pg;"[rvs off] [up]"
3410 geta$:ifa$=""then3410
3420 ifa$="[f1]"thenifpg<21thenpg=pg+1:goto3400
3430 ifa$="[f7]"thenifpg>0thenpg=pg-1:goto3400
3440 goto3100
3500 om=3
3505 geta$:ifa$=""then3500
3510 ifa$="[left]"thena=-1:gosub5300
3520 ifa$="[right]"thena=1:gosub5300
3530 ifa$="[up]"ora$="[down]"thenprint"[home]";:goto3135
3540 ifa$<>chr$(13)then3500
3550 onopgoto3600,3800,3900
3600 print"[home] [rvs on]exit[rvs off]  move pointer to entry"
3610 print"[home]";:gosub5005
3620 ifse=1then3100
3625 ifse=2then3100
3630 ad=18432+pg*92+(se-3)*4
3640 poke251,ad and 255:poke252,int(ad/256):poke253,80:poke254,4:sys12307
3650 goto3100
3800 print"[clr]are you sure you want to erase"
3805 input"everything? y[left 3]";yn$
3810 ifleft$(yn$,1)<>"y"then1100
3815 poke251,0:poke252,72:poke253,80:sys12288:goto3000
3900 goto100
4000 ifna$=""theninput"[clr]title";na$
4005 open4,4:print"[clr]printer utilities"
4010 print#4,chr$(27)+chr$(9)+chr$(20);chr$(27)+"9";spc(80);chr$(27)+"0"
4015 print"[down] print cities"
4020 print" print dots"
4030 print" print both"
4040 print" exit"
4050 print"[home][down]":mx=4:gosub5000
4060 onsegosub4100,4400,4600
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```
4070 close4:goto100
4100 cmd4:pg=1:l=16384
4120 printchr$(27)+"e";na$;" ";
4121 print"cities";chr$(27)+"r";spc(40);"page";pg:print
4125 print"designator          range                bearing":print
4130 co=1
4140 be=peek(l)+peek(l+1)*256
4145 ra=(peek(l+2)+peek(l+3)*256)/10
4150 de$="":forn=l+4tol+8:de$=de$+chr$(peek(n)):next
4151 ifleft$(de$,1)=chr$(0)then4170
4155 co=co+1:printde$;chr$(27)+chr$(9)+chr$(40);ra;chr$(27)+chr$(9)+chr$(60);be
4156 ifco=50thenco=1:pg=pg+1:l=l+9:ifl<18471thenprintchr$(12):goto4120
4160 l=l+9:ifl<18246then4140
4170 print#4,chr$(12):return
4400 cmd4:pg=1:l=18432
4420 printchr$(27)+"e";na$;" ";
4421 print"dots";chr$(27)+"r";spc(40);"page";pg:print
4425 print"range      bearing";spc(12);"range      bearing":print
4430 co=1
4440 al=l:gosub4700:printspc(16);
4445 al=l+200:gosub4700:print
4450 co=co+1
4451 ifco>50thenifep=1thenep=0:goto4470
4455 ifco>50thenco=1:pg=pg+1:l=l+204:printchr$(12):goto4420
4460 l=l+4:ifl<20479then4440
4470 print#4,chr$(12):return
4600 gosub4100:gosub4400:return
4700 ifal>20479thenreturn
4710 be=peek(al)+peek(al+1)*256:ra=(peek(al+2)+peek(al+3)*256)/10
4715 ifra=0thenep=1:return
4720 printleft$(str$(ra)+"     ",5);spc(6);left$(str$(be)+"   ",3);
4730 return
5000 se=1
5005 print"[rvs on]>[rvs off][left]";
5010 geta$:ifa$=""then5010
5020 ifa$="[up]"thenifse>1thense=se-1:print" [up][left]";
5030 ifa$="[down]"thenifse<mxthense=se+1:print" [down][left]";
5040 ifa$<>"[up]"anda$<>"[left]"thenreturn
5050 goto5005
5100 l=16384+pg*207:m=16590+pg*207
5110 de$="":forn=4to8:de$=de$+chr$(peek(l+n)):next
5120 be=peek(l)+peek(l+1)*256
5130 di=peek(l+2)+peek(l+3)*256
5140 printchr$(13);" ";de$,di/10,be;:l=l+9:ifl>mthenreturn
5150 goto5110
5200 print"[up]":printspc(sp);:open1,3
5210 get#1,b$:print"[left][rvs on]";b$;"[rvs off][left]";
5220 geta$:ifa$=""then5220
5230 ifasc(a$)<32orasc(a$)>128then5250
5240 printb$;"[left]";a$;:ifpos(0)>=le+spthenprint"[left]";
5245 goto5210
5250 ifa$="[left]"thenifpos(0)>spthen5240
5260 ifa$="[right]"then5240
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```
5270 ifa$<>chr$(13)then5220
5280 printb$;"[up]":printspc(sp);:a$=""
5290 forn=1tole:get#1,b$:a$=a$+b$:next
5295 close1:return
5300 ifop+a<lorop+a>omthenreturn
5310 gosub5350:op=op+a:gosub5375:return
5350 forn=(op-1)*7+1025 to (op-1)*7+1030:poken,peek(n)and127:next:return
5375 forn=(op-1)*7+1025 to (op-1)*7+1030:poken,peek(n)or128:next:return
5400 open3,8,0,"$0":di$(0)="":p=1:di$(p)=""
5410 forn=1to34:get#3,a$:di$(0)=di$(0)+a$:next:print"[clr]";di$(0)
5420 get#3,a$:ifst<>0then5450
5425 ifa$<>chr$(34)then5420
5430 get#3,a$:ifa$<>chr$(34)thendi$(p)=di$(p)+a$:goto5430
5440 print" ";di$(p):p=p+1:di$(p)="":goto5420
5450 close3
5460 print" exit":return
5500 open15,8,15:input#15,a,b$,c,d:close15
5505 print:printb$:print"press any key to continue"
5510 geta$:ifa$=""then5510
5520 return
5600 l=18432+pg*92:m=18523+pg*92
5620 be=peek(l)+peek(l+1)*256
5630 di=peek(l+2)+peek(l+3)*256
5640 printchr$(13);di/10,be;:l=l+4:ifl>=mthenreturn
5650 goto5620
```